

PART I - FOUNDATION PLC PROJECTS

Project 1 - Motor Start Delay System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Motor Start Delay System was the first PLC project I developed as part of my foundation work in industrial automation. The purpose of the project was to create a simple control sequence in which a motor output would not energize immediately when the start condition became active. Instead, the PLC would introduce a five-second delay before turning on the motor.

I implemented the delay using a Timer On Delay (TON) instruction. The Start and Stop input conditions controlled the timer, while the timer done bit was used separately to control the motor output. This allowed me to see clearly how a timer could be used as an intermediate control condition rather than directly controlling the physical output.

Although the application was simple, it established several concepts that became important in later projects, particularly the relationship between input conditions, timer status bits, and output control.

2. Engineering Objective

The objective of this project was to develop and test a PLC program that delayed motor energization by five seconds after the required start conditions became true. The project was also intended to give me practical experience with the TON instruction and to understand how its done status could be used elsewhere in the ladder logic to control another function.

3. Functional Requirements

- The timer shall be enabled when the Start and Stop input conditions permit operation.
- Timer T4:0 shall provide a five-second ON delay.
- The motor output shall remain OFF while the timer is timing.
- The motor output shall energize only after T4:0 reaches its preset value.
- The motor output shall de-energize when the timer-enable rung becomes false.
- The non-retentive timer shall reset when its enabling conditions are removed.

These requirements describe the actual behavior implemented in the Version 1 ladder logic.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the control program
Input Module	1762-IQ16	Provides discrete input signals
Output Module	1762-OB16	Provides discrete output control
Start Input	I:1/0	Initiates the timing condition
Stop Input	I:1/1	Controls continuation of the operating condition
Timer	T4:0	Provides the motor start delay
Motor Output	O:2/0	Represents the controlled motor command
Programming Software	RSLogix Micro Starter Lite 8.30	Ladder logic development and testing
Communication	RSLink Classic	PLC communication environment

The motor was represented through the PLC output for the purpose of this foundation project.

5. I/O Assignment

Device	PLC Address	Type	Description
Start	I:1/0	Digital Input	Enables the start-delay sequence
Stop	I:1/1	Digital Input	Provides the stop condition
Motor Output	O:2/0	Digital Output	Motor control output

Internal Data Assignment

Function	Address	Configuration	Purpose
Timer Delay	T4:0	TON	Provides the startup delay
Timer Done	T4:0/DN	Status Bit	Enables the motor after timing completes
Time Base	-	0.01 second	Timer resolution
Preset	-	500	Timer preset value

The resulting delay is calculated as: $500 \times 0.01 \text{ second} = 5.00 \text{ seconds}$.

6. Control Strategy

I used a two-stage control strategy. The first stage determines when the startup timer is allowed to operate. The Start and Stop input conditions are connected to the TON instruction T4:0. When the complete rung condition remains true, the timer begins accumulating time.

The second stage controls the motor independently from the timer rung. Instead of placing the motor output directly on the timer rung, I used the T4:0/DN status bit as the condition for O:2/0.

As a result, the motor cannot energize while the timer is still accumulating. Once the accumulated value reaches the preset of 500 at a 0.01-second time base, the done bit becomes true and the motor output is energized.

Because T4:0 is a non-retentive TON instruction, removing the timer-enable condition resets the accumulated value and clears the done status. The motor output therefore turns off when the timer condition is removed.

7. Engineering Design Decisions

Design Decision	Reason
Used a TON instruction	The application required a delay before the motor could energize.
Selected a 500 preset with a 0.01-second time base	This produced the required five-second delay.
Used T4:0/DN to control the motor	This ensured the motor could energize only after timing was complete.
Separated timer and motor logic	This made the relationship between timing and output control easier to follow.

Used a non-retentive timer	The timing sequence should restart when the timer-enable condition is removed.
Kept the Version 1 logic simple	The project focused on basic timer-based motor control before advanced functionality.

One limitation of this design is that I did not implement a separate maintained Run command or seal-in circuit. The timer-enable conditions therefore need to remain true for the timer to complete. I retained this behavior in the Version 1 documentation because it accurately represents the project I originally developed rather than a later improved version.

8. Ladder Logic Description

The Version 1 Motor Start Delay System was implemented using two functional rungs. The first rung establishes the five-second startup delay, while the second rung uses the timer completion status to control the motor output. I kept these functions separate so that the timing operation and motor-control condition could be monitored independently.

8.1 Rung 0000 - Motor Start Delay Timer

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-----[ XIC I:1/0] -----[ XIC I:1/1] -----[ TON T4:0] -----
      Start                Stop                Time Base: 0.01  Preset:   500

```

Rung 0000 controls the startup timer T4:0. When the Start I:1/0 and Stop I:1/1 input conditions make the rung true, the TON instruction begins accumulating time. If the enabling conditions are removed before the timer completes, the non-retentive TON resets and its accumulated value returns to zero.

8.2 Rung 0001 - Motor Output Control

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-----[ XIC T4:0/DN] -----(OTE O:2/0) -----
      Timer Done                Motor Output

```

Rung 0001 uses the done status of T4:0 to control Motor Output O:2/0. While the timer is accumulating, T4:0/DN remains false and the motor output remains de-energized. After the timer reaches its preset value, the done bit becomes true, making Rung 0001 true and energizing O:2/0. If the timer-enable condition becomes false, T4:0 resets, the done bit clears, and the motor output de-energizes.

8.3 Actual RSLogix Implementation

The clean ladder representations above are included to make the control sequence easier to read. The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

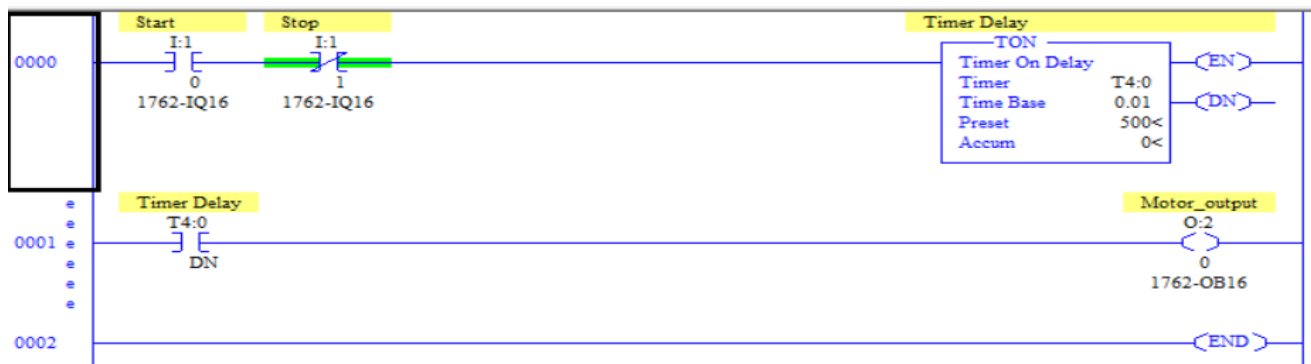


Figure P1.1 - RSLogix Implementation of the Version 1 Motor Start Delay System

The original program contains the same two control functions shown in the clean ladder representations: Rung 0000 provides the five-second delay through T4:0, and Rung 0001 uses T4:0/DN to control Motor Output O:2/0.

9. Sequence of Operation

Step 1 - Initial Condition: The motor output is OFF and T4:0 is not complete.

Step 2 - Start Condition Established: The Start and Stop input conditions make Rung 0000 true.

Step 3 - Timing Begins: T4:0 begins accumulating time while the motor remains OFF.

Step 4 - Five-Second Delay Completed: When the accumulated timer value reaches 500, the T4:0/DN bit becomes true.

Step 5 - Motor Energizes: The done bit makes Rung 0001 true and energizes O:2/0.

Step 6 - Operating Condition Removed: When the timer-enable rung becomes false, T4:0 resets.

Step 7 - Motor Stops: The timer done bit clears, causing O:2/0 to de-energize.

10. Testing & Validation

I tested the completed ladder logic in RSLogix Micro Starter Lite by monitoring the timer status, accumulated value, done bit, and motor output while applying different input conditions. Testing was performed against the functional requirements rather than relying only on visual inspection of the ladder logic.

Because the Version 1 program does not contain a maintained Run command or seal-in circuit, the Start condition must remain active for the TON instruction to continue accumulating. The tests below document the behavior of the program as it was actually implemented.

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Establish initial condition with timer-enable conditions inactive	T4:0.ACC = 0; T4:0/DN = 0; motor OFF	Timer reset and motor remained OFF	PASS
T-02	Establish Start/Stop conditions required to enable Rung 0000	T4:0 begins accumulating; motor remains OFF	Timer accumulated from zero; motor remained OFF	PASS
T-03	Maintain timer-enable condition for less than five seconds	T4:0/DN remains false; motor remains OFF	Done bit remained false; motor remained OFF	PASS
T-04	Maintain timer-enable condition for five seconds	T4:0/DN becomes true; motor energizes	Motor energized after preset was reached	PASS
T-05	Remove timer-enable condition while T4:0 is timing	Timer resets; motor remains OFF	Accumulator returned to zero; motor remained OFF	PASS
T-06	Remove timer-enable condition after motor has energized	Timer resets; Done bit clears; motor turns OFF	Done bit cleared and motor de-energized	PASS
T-07	Change timer preset during verification	Delay changes according to time base x preset	Behavior changed consistently with configured preset	PASS

Validation Summary

Testing confirmed the relationship between the timer configuration and motor-control logic. With a time base of 0.01 second and a preset of 500, T4:0 required five seconds of continuous enable before its Done bit became true.

During this timing period, O:2/0 remained de-energized. Once the preset was reached, T4:0/DN enabled the motor-output rung. Removing the timer-enable condition reset the non-retentive timer, cleared its Done bit, and removed the motor command.

The additional preset test also confirmed that the delay was determined by the relationship between the timer time base and preset rather than by the preset value alone.

These tests provided functional validation of the Version 1 program within the project environment. They do not represent industrial commissioning or safety validation.

11. Results

The completed program successfully produced a five-second delay between establishing the required start condition and energizing the motor output. The TON instruction accumulated to its preset value, after which its done bit enabled the motor output.

The project also confirmed that removing the timer-enable condition reset the non-retentive timer and de-energized the motor. The functional requirements established for this foundation implementation were therefore satisfied during testing.

12. Challenges & Troubleshooting

This project helped me develop a more systematic approach to PLC troubleshooting. Because the program contained only two functional rungs, I could monitor each condition individually and observe how a problem in the timer rung affected the motor-output rung.

Instead of changing several instructions when the sequence did not behave as expected, I checked the relevant input conditions, timer accumulator, timer Done bit, and motor-output condition in sequence.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Timer does not start	Start/Stop conditions are not making Rung 0000 true	Monitor I:1/0, I:1/1, and rung continuity	Verify input states and instruction/contact selection
Timer starts but resets before five seconds	Timer-enable condition does not remain true	Monitor Rung 0000 and T4:0.ACC	Verify required input conditions remain active
Motor starts immediately	Wrong condition or timer status controls output	Inspect Rung 0001	Ensure T4:0/DN controls O:2/0
Motor never starts	Timer does not reach preset or Done bit is misaddressed	Monitor T4:0.ACC and T4:0/DN	Verify timer configuration and status-bit address
Motor does not turn OFF when timer resets	Output logic is not responding to Done-bit state	Monitor T4:0/DN and O:2/0	Verify motor rung and output address
Incorrect startup delay	Preset interpreted without considering time base	Check T4:0 configuration	Calculate delay using preset x time base
Instruction-entry error	Instruction or address entered incorrectly	Review affected rung in RSLogix	Insert correct instruction and intended address

One of the most useful lessons from troubleshooting this project was the importance of following the signal path through the program: Input conditions -> Timer enable -> Timer accumulator -> Timer Done bit -> Motor output.

I also learned to verify the timer configuration numerically. A preset of 500 does not by itself mean five seconds. With the configured 0.01-second time base, $500 \times 0.01 \text{ second} = 5.00 \text{ seconds}$. Monitoring the program while it was running made these relationships much clearer than reviewing the ladder logic offline alone.

13. Lessons Learned

This project gave me my first practical understanding of using a timer as part of a control sequence rather than simply treating it as an isolated PLC instruction.

I learned how the timer preset and time base determine the actual delay, how the .DN status bit can be used as a condition elsewhere in the program, and how a non-retentive timer responds when its enabling rung becomes false.

I also learned that even simple ladder logic benefits from separating the condition that performs the timing from the logic that controls the final output. This became more useful as I moved into projects with multiple timers, counters, and sequential operations.

14. Industrial Relevance

Delayed equipment startup is used in many industrial control applications. A delay may be required to establish process conditions, coordinate the startup of multiple pieces of equipment, allow supporting equipment to become ready, or prevent several loads from starting simultaneously.

The basic timer-controlled output developed in this project represents the foundation of those applications. However, an actual industrial motor-control system would normally require additional hardware and control considerations, including appropriate motor starters or drives, overload protection, emergency stopping functions, operating permissives, fault handling, and applicable safety requirements.

For that reason, this Version 1 program should be viewed as a foundation control exercise rather than a production-ready motor control design.

15. Transition to Industrial Version (Version 2)

The Version 1 Motor Start Delay project established the basic relationship between an operating condition, a TON timer, and a delayed motor output. It intentionally remained a simple foundation program. The Version 2 project will retain the basic five-second startup-delay requirement while redesigning the control system using stronger industrial control practices.

Version 2 Improvement	Engineering Benefit
Maintained Run Request using an internal control bit	Allows a momentary Start command to establish a maintained operating request until Stop is activated
Auto/Manual operating modes	Supports normal automatic operation as well as controlled maintenance or commissioning operation
Mode-conflict detection	Identifies an invalid condition if Auto and Manual commands are active simultaneously
Start permissives and interlocks	Prevents startup unless required operating conditions are satisfied
Motor overload feedback	Prevents or removes the run command when an overload

	condition exists and provides fault information
Pre-start warning	Provides an indication before equipment motion begins
Fault and alarm handling	Provides defined responses to abnormal operating conditions
HMI-adjustable startup delay	Allows authorized adjustment of the delay and provides timer/status information to the operator
Improved status and diagnostics	Makes Run Request, timer status, motor status, mode, permissives, and faults easier to troubleshoot
Safety-system interface considerations	Separates standard PLC control functions from safety-related shutdown functions where applicable

A major improvement will be the introduction of a maintained Run Request. In Version 1, the timer-enable conditions must remain true for T4:0 to complete its five-second timing period. Version 2 can separate the operator's Start command from the maintained internal request.

Start Request -> Run Request -> Mode / Permissive / Interlock Checks -> Pre-start Delay -> Motor Command -> Running Status / Diagnostics

Fault and Stop conditions can then remove or inhibit the Run Request according to the final Version 2 control philosophy. This creates a meaningful engineering progression between the two implementations. Version 1 demonstrates the fundamental PLC timing concept, while Version 2 will show how that concept can be incorporated into a more complete industrial control architecture.

16. Project Summary

I developed the Motor Start Delay System to build a practical understanding of timer-based PLC control. The program used the Start and Stop input conditions to enable a TON instruction and used the timer done bit to energize the motor output after a five-second delay.

Although the ladder logic was simple, the project established several concepts that I used repeatedly in later work, including timer configuration, status-bit evaluation, separation of control functions, and systematic testing.

The project also provided a clear starting point for future improvement. Its limitations, particularly the absence of a maintained Run command and industrial motor permissives, provide specific engineering objectives that can be addressed when the system is redesigned as the Version 2 industrial project.